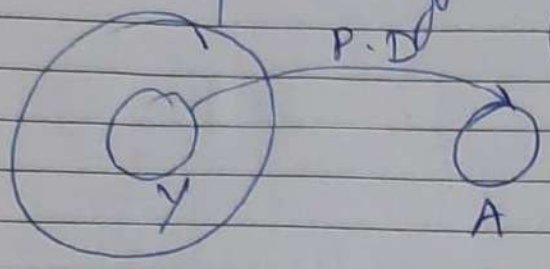
2NF: A relation is in 2NF iff it is in 1NF & all the non-key attributes are fully functional dependent on the primary key.



X : CK
A : NPA
Y C X

(OR)

- R should be in 1NF
- And, R should not contain any partial dependency.

⇒ $Y \rightarrow A$ is partially functional dependent if,

$((Y \subset CK) \wedge (A : NPA))$

⇒ $Y \rightarrow A$ is fully functional dependent if,

$((Y \not\subset CK) \vee (A : PA))$

Ex1: R (A B C D)

if, CK → AB

Prime attributes = {A, B}

Non Prime Attributes = {C, D}

Life is pleasant. Death is peaceful. It's the transition that's troublesome.

→ A → CD (P.F.D)
(as, A ⊂ CK and CD both are NPA)
→ B → D (P.F.D)
(as, B ⊂ CK and D is a NPA)
→ B → AC
B → A (F.F.D) (∵ A is a PA)
B → C (P.F.D) (∵ B ⊂ CK and C is a NPA)

* Note: If the P.K consists of compound or composite attribute, then go for 2NF decomposition.

Ex 2: R(ABCDE)

F = {AB → C, C → D, B → E}

CK = {AB} ∵ AB^+ → {ABCDE}

Prime attributes → {A, B}

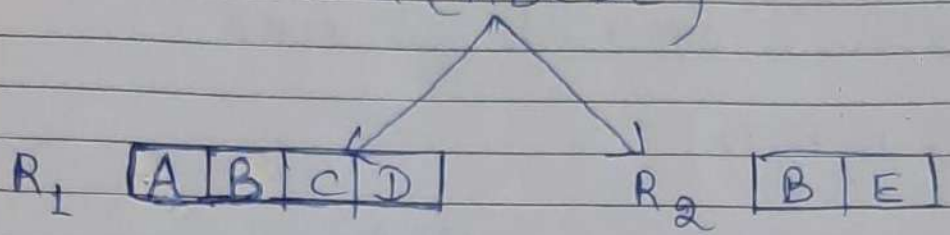
Non-prime attributes → {C, D, E}

AB → C (F.F.D)
C → D (P.F.D)
B → E (P.F.D)

S M T W T F S
1 2 3
4 5 6 7 8 9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

2NF decomposition:

R(ABCDE)



$AB \rightarrow C$ (FFD)
$C \rightarrow D$ (FFD)
⇒ in 1NF
⇒ all NK attributes are FFD on P.K.
(2NF)

$B \rightarrow E$
⇒ in 1NF
⇒ Non-key attribute E is fully functional dependent on PK B.
(2NF)

3NF: A relation is in 3NF iff it is in 2NF & there exists no transitive dependency in the relation.

$X \rightarrow Y$ is transitive dependent if,

$[(X \text{ is not a S.K}) \cap (Y : \text{N.P.A})]$

* **Note:** If a NK → NK is present, then go for 3NF decomposition

Ex1: R(ABCDEF)

if CK → {AB, CD}

Prime → {A, B, C, D}
Non-prime → {E, F}

First they ignore you, then they laugh at you, then they fight you, then you win

$ABE \rightarrow F$
$A \rightarrow E$ (T.D) (∵ A is not a S.K & E is a NPA):
$A \rightarrow B$

Ex2: R(ABCDE)

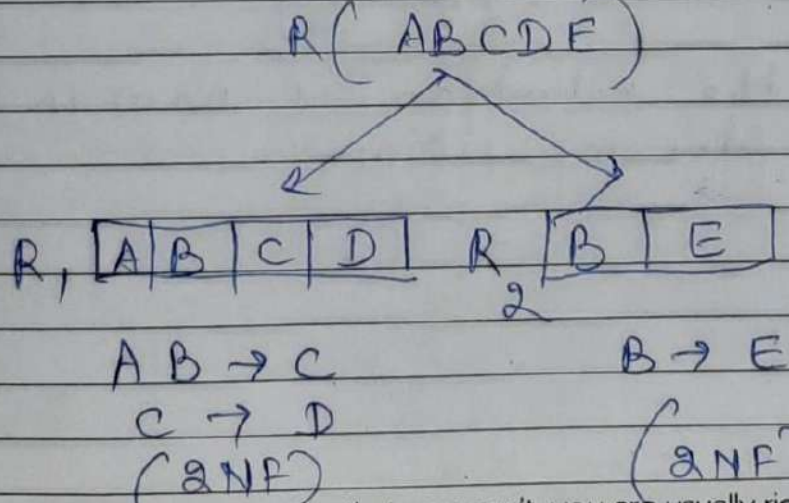
$F = \{AB \rightarrow C, C \rightarrow D, B \rightarrow E\}$

Check whether the relation is in 3NF or not. If not, then decompose it into 3NF

A. $AB^+ = \{ABCDE\}$

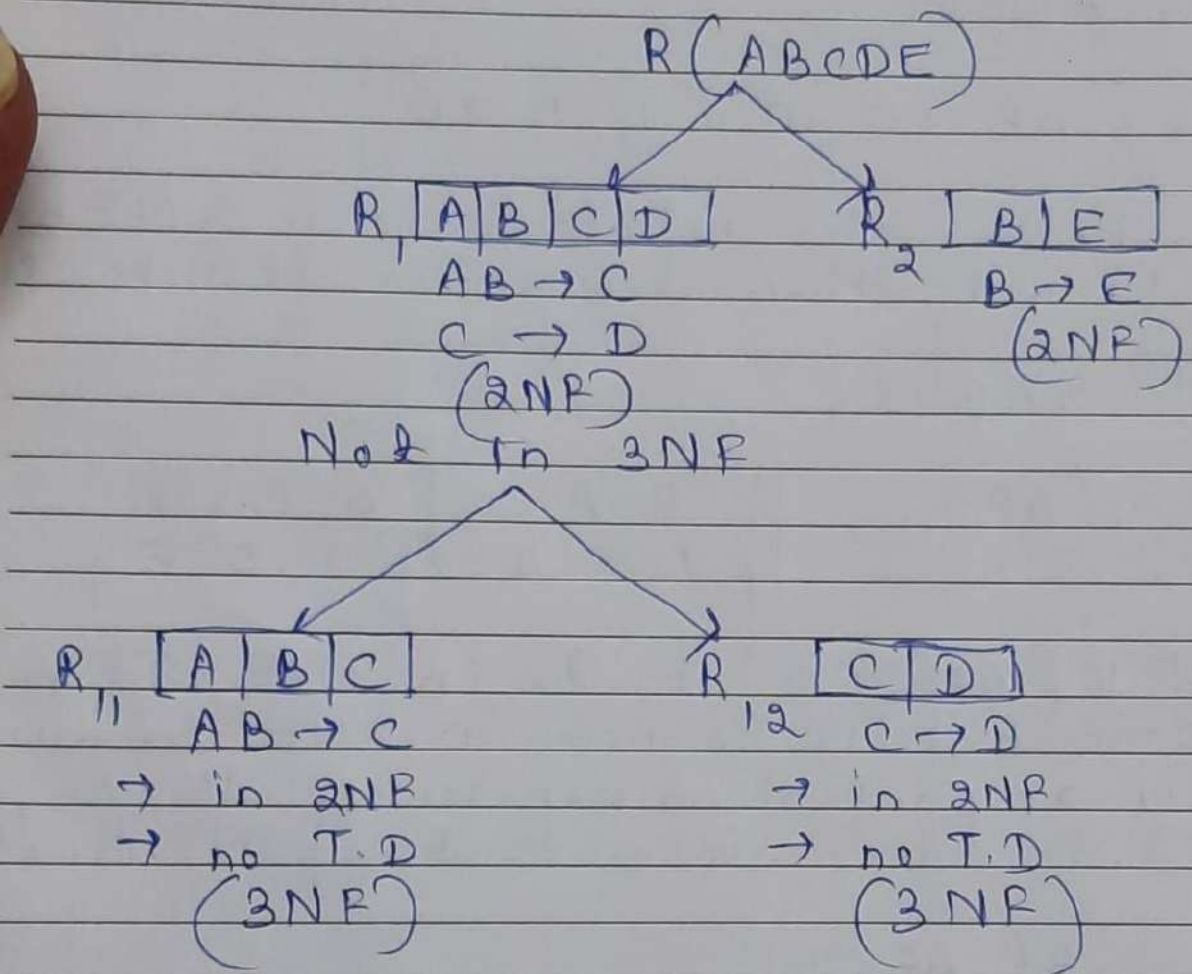
C.K $\rightarrow \{AB\}$ || P.A $\rightarrow \{A, B\}$
N.P.A $\rightarrow \{C, D, E\}$

→ For rela^n, R(ABCDE) to be in 3NF, It must be in 2NF. But it is not in 2NF because $B \rightarrow E$ is a partial dependency. Hence, we need to decompose it to 2NF first.



S	M	T	W	T	F	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

Now, the relation is in 2NF. But still it is not in 3NF because C → D has transitive dependency. Hence, we need to decompose it to 3NF.



11 SUNDAY Hence, the relation is now in 3NF after decomposition.

He who has a 'why' to live, can bear with almost any 'how'.

BCNF (Boyce Codd Normal Form (or) 3.5 NF)

A relation is said to be in BCNF iff
- it should be in 3NF
- for any dependency $A \to B$, A should be a super key.

which means,
for, $A \to B$
if A is non-prime and B is a prime attribute, then not in BCNF.

* **Note**: If NK → Key exists, then go for BCNF decomposition.

BCNF decomposition can be done when:
- there are multiple candidate keys
- all the C.Ks are composite in nature
- all the C.Ks are overlapping in nature

Ex: R(ABCD)

$F = \{AB \to CD, AC \to BD, D \to B\}$

A. $AB^+ = ABCD$
$AC^+ = ACBD$
$D^+ = DB$

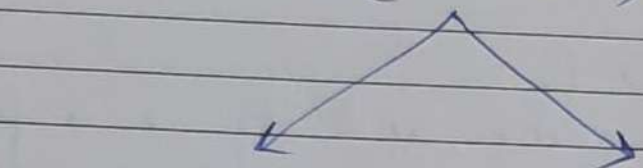
C.Ks = {AB, AC}

MAR 2012
S M T W T F S
1 2 3
4 5 6 7 8 9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

Prime attributes → {A, B, C}
Non-prime/Non-key attributes → D

As, D → B is present
i.e NK → K is present, the relation is not in BCNF. We can go for decomposition.

R(ABCD)



R_1 | D | B

D → B
(∵ $D^+ = DB$)
(BCNF)

R_2 | A | C | D

AC → D
(∵ $AC^+ = ACD$)
(BCNF)

You can only find truth with logic if you have already found truth without it.

4NF :

A relation will be in 4NF if it is in BCNF and has no non-trivial multi-valued dependency

→ $A \twoheadrightarrow B$, is multivalued dependency if

A1 < B1, B2

→ A table should have atleast 3 columns to have MVD.

→ For a table with A, B, C columns, B & C should be independent of each other.

Ex: Enrolment table → No relationship (Bad design)

Sid	Course	Hobby
1	Science	Cricket
1	Maths	Hockey
2	C++	Cricket
2	PHP	Hockey

Course_Opted table

Sid	Course
1	Science
1	Maths
2	C++
2	PHP

Hobbies table

Sid	Hobby
1	Cricket
1	Hockey
2	Cricket
2	Hockey

4NF

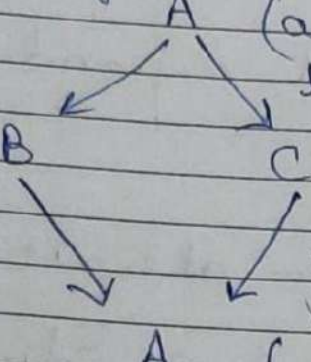
∴ (SID $\twoheadrightarrow$ course [trivial MVD] & SID $\twoheadrightarrow$ Hobby (trivial MVD)) 2 independent tables (Good design)

* trivial : {SID} ∪ {Course} = course_opted table

MAR 2012
S M T W T F S
1 2 3
4 5 6 7 8 9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

5NF : (PJNF) Project Join Normal Form

A relation is in 5NF if it is in 4NF & not contains any join dependency and joining should be lossless.



Ex:

SPC table

Supplier	Product	Customer
Acme	72XSW	Ford
Acme	Gear L	GM
Robuto	E Switch	Ford
Robuto	OBD II	Mercedes
Alwat	72XSW	GM
Alwat	OBD II	Mercedes
Alwat	Gear L	Mercedes

Primary Key :

(# supplier, # product, # Customer) → Composite Primary Key

Rules for checking lossless join decompos^n (for binary decomposition)

If R is decomposed into R_1 & R_2, then one of the following cond^n holds:

i) $R_1 \cap R_2 \rightarrow R_1 - R_2$ is in F+

(OR)

ii) $R_1 \cap R_2 \rightarrow R_2 - R_1$ is in F+

→ if either of the rel^n holds, then it is lossless

Ex1: R(A, B, C, D)

$F = \{AB \rightarrow C, C \rightarrow D\}$

Suppose R is decomposed into two rel^n $R_1(A,B,C)$ and $R_2(C,D)$.

A: $R_1 \cap R_2 \rightarrow R_2 - R_1$

$\Rightarrow C \rightarrow D$

As, $C \rightarrow D$ is present in F+

(or)

as, $C \rightarrow D \in F^+$, so it is lossless.

Ex2: Suppose R decomposed into R_1 & R_2

R(C, S, Z)

$R_1(S, Z)$ and $R_2(C, Z)$.

$F = \{CS \rightarrow Z, Z \rightarrow C\}$ Test for lossless join decomposition and dependency preserving decomposition.

17 MARCH

MAR 2012
S M T W T F S
1 2 3
4 5 6 7 8 9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

12TH WEEK 77 DAYS

4NF (Contd)

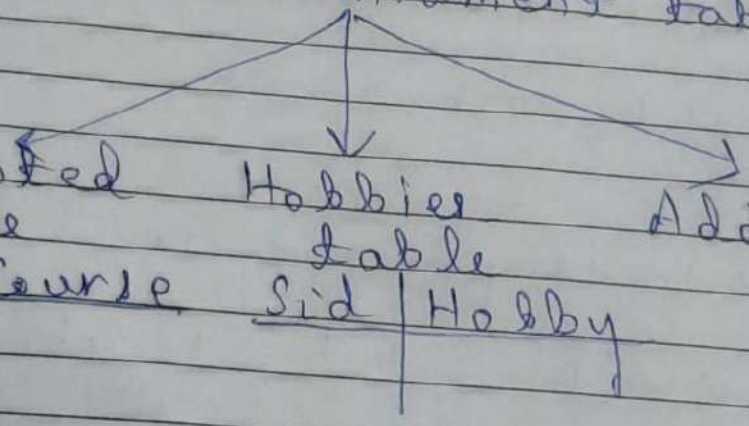
A table can have both FD & MVD. Let's add a new column address to our original Enrolment table.

Sid	Address	Course	Hobby

Sid → Address (FD)
Sid →→ Course (MVD)
Sid →→ Hobby (MVD)

In this case, the decomposition will be,

Student Enrolment table



Course_Opted table

Sid	Course

Hobbies table

Sid	Hobby

Address table

Sid	Address

18 SUNDAY

⇒ Hence, MVD occurs due to bad database design.

⇒ A Good database design can provide the solution.

2012

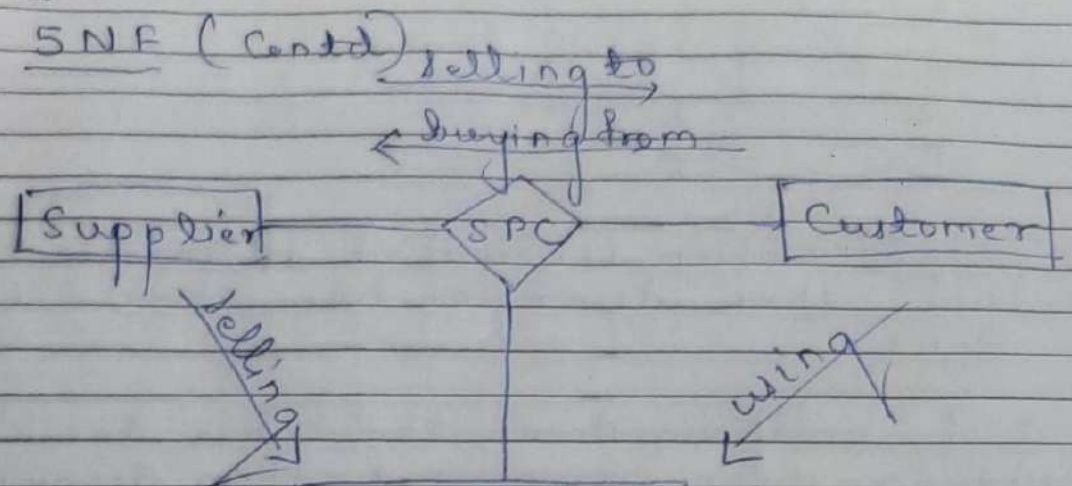
The secret of success is to know something nobody else knows.

APR 2012
S M T W T F S
1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

13TH WEEK 79 DAYS

MONDAY 19 MARCH

5NF (Contd)



(Original table)

SPC Table

Supplier	Product	Customer

SP table

Supplier	Product

(Table 1)

SC table

Supplier	Customer

(Table 2)

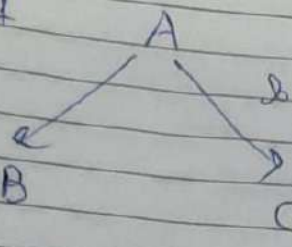
CP table

Customer	Product

(Table 3)

MAR 2012
S M T W T F S
4 5 6 7 8 9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

→ if



break down

Add'l information is created or information is lost, then decomposition should not be done.

→ But if breaking down the table doesn't lead to information loss, then decomposition is allowed.

** Decomposition is of 2 types:

→ From the diff. tables/relations in the given example, if we perform natural join between any of the two relations i.e. SP ⋈ SC, SC ⋈ CP or SP ⋈ CP, then extra rows are added ⇒ this decomposition is called lossy decomposition.

→ But if we perform natural join between the above three relations, then no extra rows are added ⇒ this decomposition is called lossless decomposition.

⇓

(Then, the above 3 tables/relations are in 5NF)

The artist is nothing without the gift, but the gift is nothing without work.

Q. Example of lossless join decomposition → (Non-additive) ID

Employee – Department table:

Eid	EName	EAge	ECity	Did	DName
22	Denim	28	Mumbai	827	Sales
33	Elina	25	Delhi	438	Marketing
46	Stephen	30	Bangalore	869	Finance
52	Katherine	36	Mumbai	575	Production
60	Jack	40	Noida	678	Testing

Employee table:

Eid	EName	EAge	ECity
22	Denim	28	Mumbai
33	Elina	25	Delhi
46	Stephen	30	Bangalore
52	Katherine	36	Mumbai
60	Jack	40	Noida

Department table:

Did	Eid	DName
827	22	Sales
438	33	Marketing
869	46	Finance
575	52	Production
678	60	Testing

(*) → Natural join

Employee (⋈) Department

Eid	EName	EAge	ECity	Did	DName
22	Denim	28	Mumbai	827	Sales
33	Elina	25	Delhi	438	Marketing
46	Stephen	30	Bangalore	869	Finance
52	Katherine	36	Mumbai	575	Production
60	Jack	40	Noida	678	Testing

→ (Original Relation → hence, lossless)

MAR 2012
S M T W T F S
1 2 3
4 5 6 7 8 9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

Q. Given a Relation R = (A B C D E). We decompose it into R_1 = (A, B, C) and R_2 = (C, D, E). Show that this decomposition is not a lossless join decomposition or lossy decomposition.

R

A	B	C	D	E
1	1	d	1	1
2	2	d	2	2

R_1

A	B	C
1	1	d
2	2	d

R_2

C	D	E
d	1	1
d	2	2

$\Pi_{A,B,C}(R_1) * \Pi_{C,D,E}(R_2)$

A	B	C	D	E
1	1	d	1	1
1	1	d	2	2
2	2	d	1	1
2	2	d	2	2

→ extra rows are added. Hence, it is a lossy decomposition.

Obstacles are those frightful things you see when you take your eyes off your goal.

Dependency Preserving Decomposition

A Dependency Preserving Decomposition of a relation R is R1, R2, R3.....Rn concerning the set of functional dependencies FD if,

$$(FD1 \cup FD2 \cup ..... \cup FDn)^+ = FD^+$$

where,

- FD1, FD2, FD3....FDn sets of functional dependencies of relations R1, R2, R3....Rn.
- $(FD1 \cup FD2 \cup FD3 \cup .... FDn)^+$ → closure of union of all sets of functional dependencies
- FD^+ → closure of set of FDs of R.

⇒ if R decomposed to R1 with FD1 & R2 with FD2, then

i) $FD1 \cup FD2 = FD$ (Dependency Preserving)

ii) $FD1 \cup FD2$ is a subset of FD (Not Dependency Preserving)

iii) $FD1 \cup FD2$ is a superset of FD (Not possible)

Example: Let the relation R(A, B, C, D) & functional dependency {AB → C, C → D, D → A}. R is decomposed into R_1(A, B, C) & R_2(C, D). check whether the decomposition is dependency preserving or not.

A- R_1(A, B, C)

R_2(C, D)

MAR 2012
S M T W T F S
1 2 3
4 5 6 7 8 9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

Let us find closure of F_1 & F_2.
To find closure of F_1, Consider all combinations of ABC. i.e, find closure of A, B, C, AB, BC and AC.
ABC can't be considered as it will always yield ABC.

$A^+ = A$ (trivial)
$B^+ = B$ (trivial)
$C^+ = (C, A, D)$
→ D can't be considered as it is not present in R_1
→ C will be removed as trivial,
Hence, $C \rightarrow A$ is included in F_1.

$AB^+ = (A, B, C, D)$
→ AB is trivial, so will be removed.
→ D can't be considered as it is not present in R_1.
Hence, $AB \rightarrow C$ is included in F_1.

SUNDAY

$BC^+ = (B, C, D, A)$
→ BC is trivial, so removed.
→ D not considered as not present in R_1.
Hence, $BC \rightarrow A$ is included in F_1.

$AC^+ = (A, C, D)$
→ AC will be removed as trivial
→ D not considered as not present in R_1.

Life is pleasant. Death is peaceful. It's the transition that's troublesome.

Hence, $F_1 = \{C \to A, AB \to C, BC \to A\}$

To find closure of F_2, consider all combination of CD i.e C, D, CD.

$C^+ = (C, D, A)$ ⟶ C is trivial, so removed; A is not considered as not present in R_2

$D^+ = D$ (trivial)

$CD^+ = (C, D, A)$ (same as above)

Hence, $C \to D$ is included in R_2.

Hence, $F_2 = \{C \to D\}$.

In the original relation,

Dependency = $\{AB \to C, C \to D, D \to A\}$

$AB \to C$ is present in F_1,
$C \to D$ is present in F_2
$D \to A$ is not preserved.

⇒ $F_1 \cup F_2$ is a subset of F. So, decomposition is not dependency preserving.

Q- R(ABC), $F = \{A \to B, B \to C\}$
Decomposition of R: $R_1(A, B)$ & $R_2(B, C)$
Does this decomposition preserve the given dependencies?